

Kubernetes CronJob Controller

avengeHERs

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Here's the team!

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Idea

We aim to develop a custom Kubernetes CronJob controller using Go.

- Objective: Develop a basic custom Kubernetes controller using Go.
- Purpose: Learn and understand the fundamentals of Kubernetes and its controller development.



Figure: Kubernetes

What is Kubernetes (K8s)?

- It is a container orchestration platform that helps in automating the deployment, scaling, and management of containerized applications.
- Think of Kubernetes as a manager for your application containers, ensuring they run smoothly, are scaled correctly, and recover from failures.

K8s: Deploy, Scale, Chill.



Why K8s?

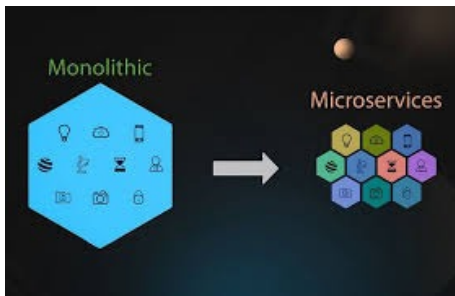
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- Trend from Monolith to Microservices.
- Increased usage of containers.
- Demand for a proper way of managing those hundreds of containers.



Inside K8s : The role of Controllers

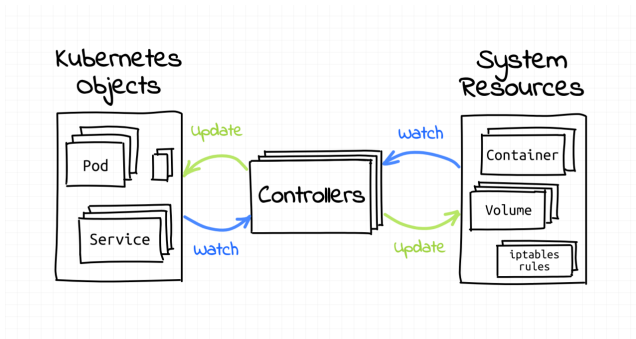
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Inside K8s : The role of Controllers

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The CronJob Controller

A CronJob controller in Kubernetes is like a scheduler or a task manager for your applications running in a Kubernetes cluster. It allows you to automate repetitive tasks that need to be performed at specific times or intervals.

Tool Chain

- Programming Language: Go (Golang)
- Kubernetes Client Library: client-go
- Controller Framework: controller-runtime
- Development Tools: Go tools (go), go mod
- Testing Framework: Go testing package, controller-runtime testing utilities
- Containerization: Docker
- KubeBuilder Framework
- Version Control: Git

Plan of action

- Learn Go
- Learn Kubernetes concepts
- Environment Setup
- Design and Implementation
 - Design Custom Resource Definitions (CRDs)
 - Develop CronJob Controller
 - Integrate with Kubernetes API
 - Update the mentors regarding the progress after a week , take feedback and work accordingly.
- Testing and Quality Assurance
- Deployment

Resources

- Kubernetes Documentation - <https://kubernetes.io/>
- Go Documentation - <https://go.dev/doc/>

Thank you

```
fmt.Println(" Thank you")
```